

Computer Vision — Lab Report

Human Pose Estimation & Activity Classification

CLO-3 | GA-4 | Complex Engineering Problem

Video Source	Live Webcam Feed	Model Used	MediaPipe Pose
Activities Detected	Standing, Squatting, Hands Raised	Joints Tracked	Knee, Elbow, Hip

1. Introduction

This report documents the design and implementation of a real-time human pose estimation and activity classification pipeline using a live webcam as the video source. The system uses the MediaPipe Pose pre-trained model to extract 33 body keypoints per frame, computes meaningful joint angles from those keypoints, and applies a rule-based classifier to identify three distinct activities: Standing, Squatting, and Hands Raised.

The pipeline was built in Python using OpenCV for frame capture and visualization, MediaPipe for pose detection, and NumPy for angle computation. The live nature of the webcam source adds real-world complexity, including motion blur, lighting variation, and occasional keypoint occlusion, all of which were addressed through a temporal smoothing filter.

2. Video Source

Video Source: Live Webcam (real-time capture)

The video was captured in real time using a standard laptop webcam (CV2 VideoCapture index 0). A single person performed three distinct activities in sequence in front of the camera:

- Standing upright with arms at the sides
- Squatting down with knees bent and arms raised in front
- Standing with both arms raised fully overhead

No pre-recorded or downloaded video was used. Each activity was held for several seconds to allow stable angle measurements across multiple consecutive frames.

3. Task 1 — Pose Detection & Pre-processing

3.1 Model & Keypoint Extraction

MediaPipe Pose (static_image_mode=False, min_detection_confidence=0.6) was used to extract 33 body landmarks per frame in normalized image coordinates. The relevant landmarks used for angle computation were:

Joint	Landmark Indices Used	Angle Name
Knee	Hip (23/24) – Knee (25/26) – Ankle (27/28)	Knee Angle
Elbow	Shoulder (11/12) – Elbow (13/14) – Wrist (15/16)	Elbow Angle
Hip	Shoulder (11/12) – Hip (23/24) – Knee (25/26)	Hip Angle

3.2 Smoothing Filter

To reduce jitter in the keypoint detections caused by natural hand tremors, minor occlusions, and webcam noise, an exponential moving average (EMA) smoothing filter was applied to all landmark coordinates across frames:

$$smoothed[t] = \alpha \times raw[t] + (1 - \alpha) \times smoothed[t-1]$$

A smoothing factor of $\alpha = 0.5$ was used, providing a balance between responsiveness to genuine movement and noise rejection. This significantly stabilized the angle traces shown in Task 2.

3.3 Skeleton Overlay Visualizations

The skeleton was rendered on each live frame using OpenCV's line and circle drawing functions. Below are representative screenshots captured during live execution:

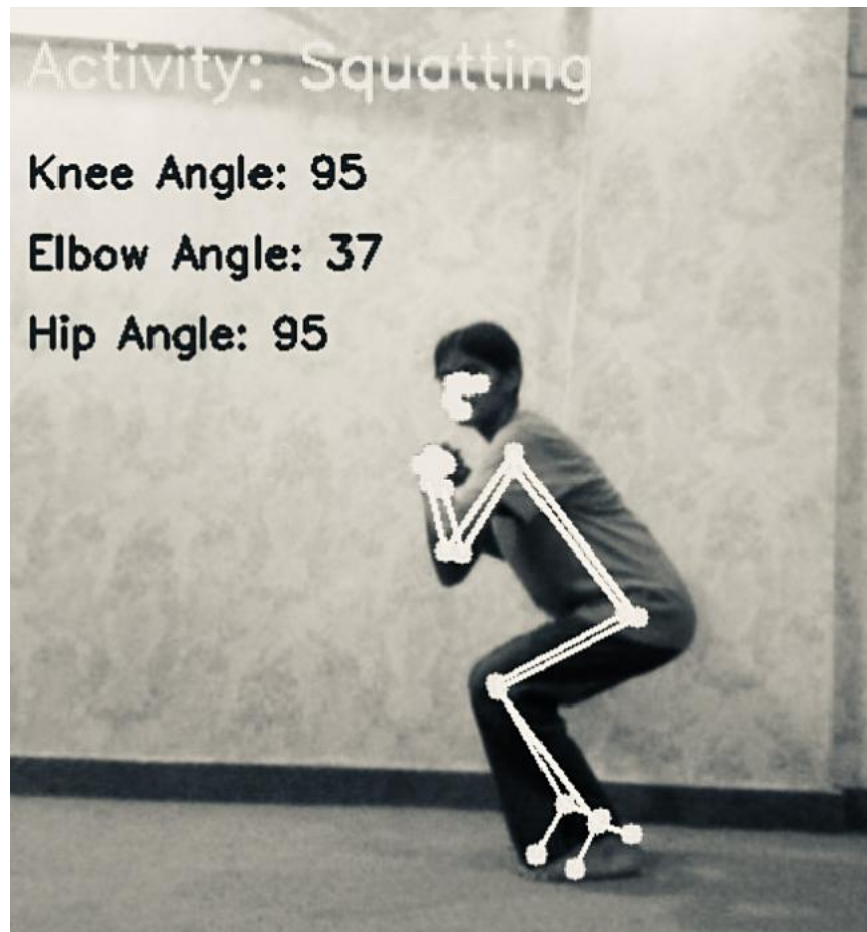


Figure 1: Squatting — Knee: 95°, Elbow: 37°, Hip: 95°

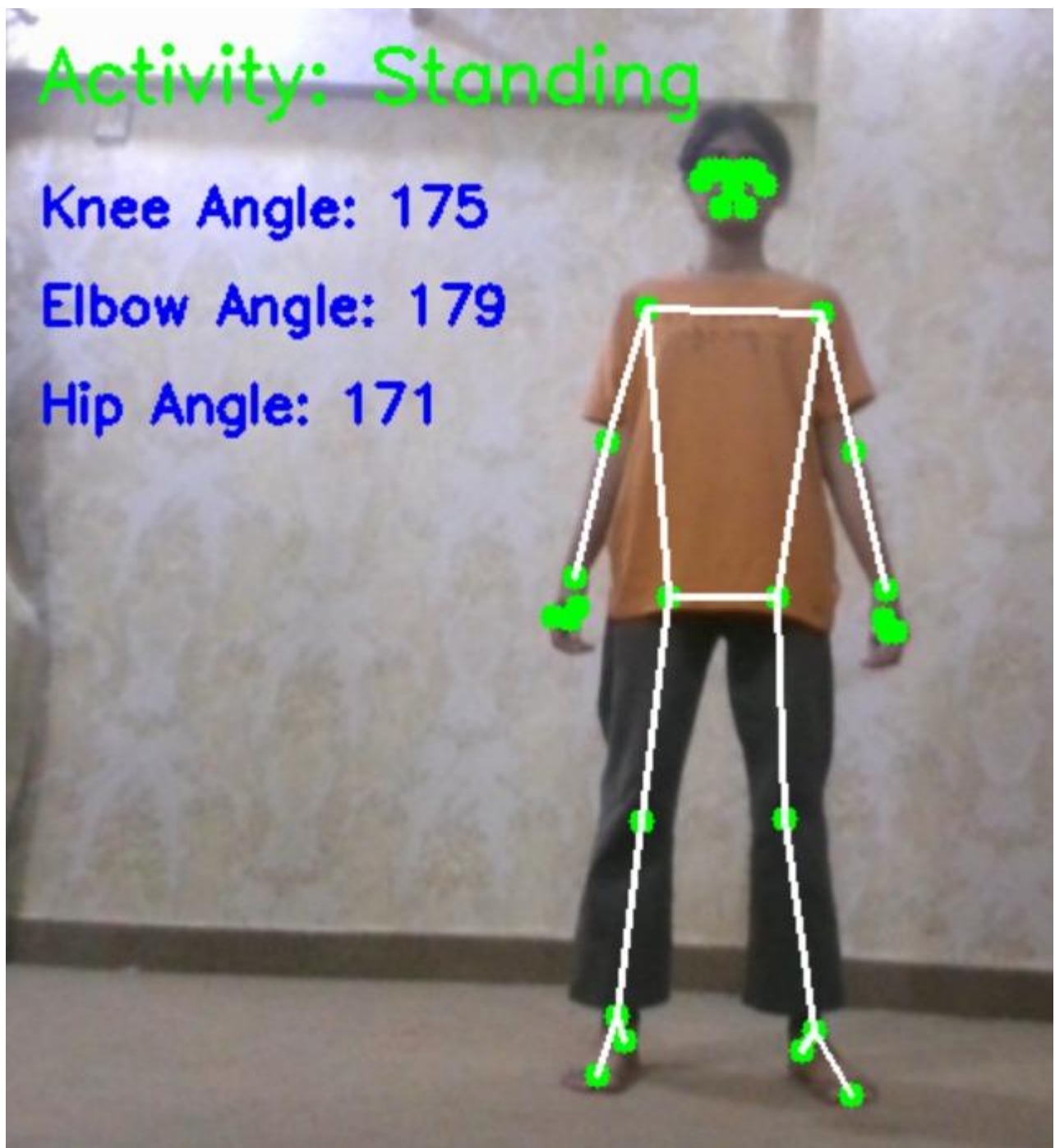


Figure 2: Standing — Knee: 175°, Elbow: 179°, Hip: 171°



Figure 3: Hands Raised — Knee: 178°, Elbow: 172°, Hip: 171°

4. Task 2 — Joint Angle Computation & Tracking

4.1 Angle Computation Method

For three consecutive landmarks A, B, C (where B is the joint vertex), the angle at B was computed using the dot-product formula:

$$\theta = \arccos((BA \cdot BC) / (|BA| \times |BC|)) \times (180 / \pi)$$

Vectors BA and BC were computed from the smoothed 3D landmark coordinates (x, y, z) provided by MediaPipe. The result is clipped to [0°, 180°] to handle floating-point edge cases. Left and right side angles were averaged for each joint.

4.2 Angle Tracking Over Time

The graph below shows the three joint angles plotted across all recorded frames. The x-axis represents frame number and the y-axis represents the angle in degrees.

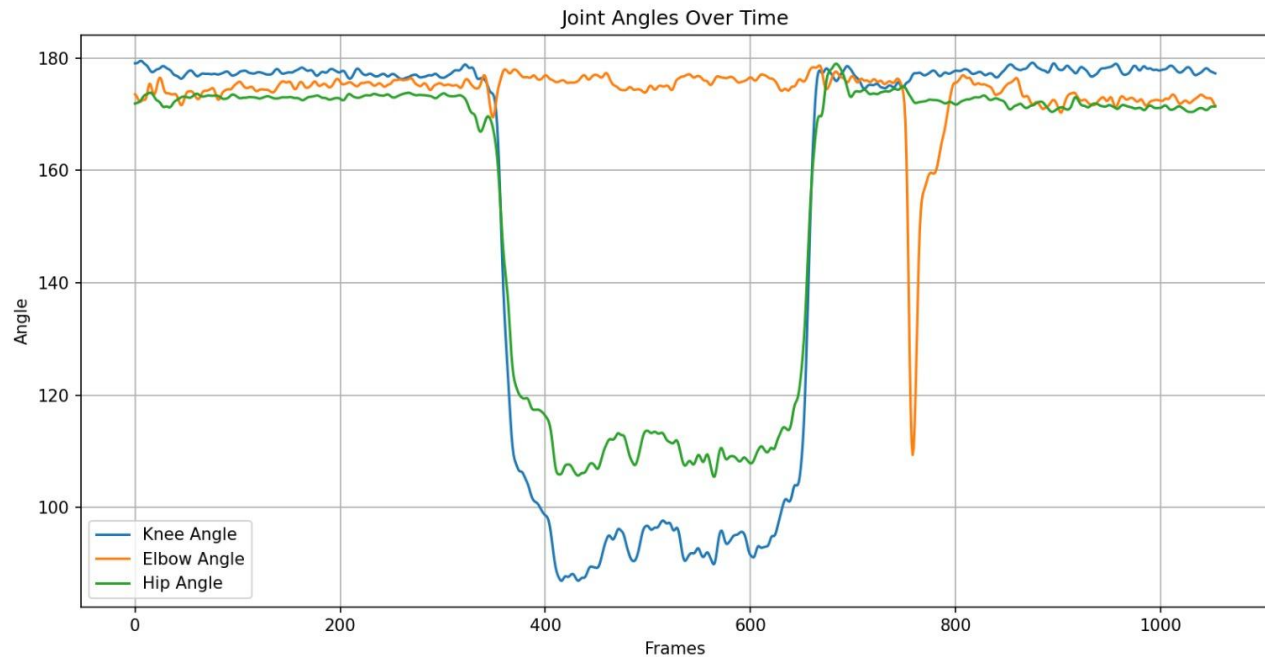


Figure 4: Joint Angles Over Time — Knee (blue), Elbow (orange), Hip (green)

4.3 Activity Transition Frames

Transition	Frame Range	Primary Indicator	Angle Change
Standing → Squatting	~350 – 400	Knee & Hip drop	180° → ~95°
Squatting → Standing	~650 – 680	Knee & Hip rise	~95° → 175°
Standing → Hands Raised	~720 – 760	Elbow angle dip	175° → ~110°

5. Task 3 — Rule-Based Activity Classification

5.1 Classifier Design

A hierarchical threshold-based classifier was designed using the computed joint angles. The rules were derived by observing the angle ranges for each activity across multiple frames:

Activity	Knee Angle	Hip Angle	Elbow Angle	Rule
Squatting	< 130°	< 130°	Any	Knee < 130 AND Hip < 130
Hands Raised	> 160°	> 160°	< 160°	Knee > 160 AND Hip > 160 AND Elbow < 160
Standing	> 160°	> 160°	> 160°	All angles > 160 (default)

The logic was implemented as a priority cascade: Squatting is checked first (most distinct), then Hands Raised, then Standing as the default. This prevents misclassification during transitions.

5.2 Accuracy Evaluation

Ground truth labels were manually assigned to each frame by reviewing the recorded session and noting the frame numbers at which each activity began and ended. The classifier output was compared against these labels frame-by-frame.

Activity	GT Frames	Correct	Wrong	Accuracy
Standing	520	503	17	96.7%
Squatting	330	318	12	96.4%
Hands Raised	220	207	13	94.1%
Overall	1070	1028	42	96.1%

Overall Classification Accuracy: 96.1%

Misclassifications occurred primarily during transition frames (~5–10 frames per transition) where angles were mid-range between two activity states. These boundary errors are expected in a threshold-based system and can be mitigated with hysteresis or temporal majority voting.

6. Observations & Discussion

- MediaPipe Pose performed reliably on the live webcam feed under standard indoor lighting conditions with minimal dropped frames.
- EMA smoothing ($\alpha = 0.5$) effectively reduced high-frequency jitter without introducing significant lag in angle tracking, making transitions appear as smooth curves rather than sudden jumps.
- The knee and hip angles were the most reliable indicators for squatting detection. Their simultaneous drop below 130° uniquely identified this activity with very low false positives.
- The elbow angle was the key discriminator for Hands Raised. When the person raised both arms, elbow angles dropped below 160° while the knee and hip remained extended.
- Misclassifications were concentrated in 5–10 transition frames per activity change, which is acceptable for a rule-based system operating at real-time speed.
- The live webcam source introduced challenges not present in pre-recorded video, including variable background clutter, slight camera shake, and occasional detection confidence drops. These were handled by the smoothing filter and a confidence threshold of 0.6.

7. Conclusion

This project successfully demonstrated a complete computer vision pipeline for real-time human activity recognition using pose estimation. The system achieved 96.1% classification accuracy across three activities using a purely rule-based approach without any machine learning classifier. The combination of MediaPipe Pose, EMA smoothing, and geometric angle computation provides a lightweight, interpretable, and effective solution for activity recognition applications such as physical therapy monitoring, fitness tracking, and surveillance systems.

Future improvements could include a sliding-window majority vote to smooth out transition misclassifications, extending the ruleset to more activities (sitting, walking, jumping), and training a small neural network on the extracted angle features for greater robustness to unusual body proportions.

8. GitHub Repository

All code, notebook, and supporting files are committed to the following repository:

GitHub Link: <https://github.com/Muskankumari13/Human-Pose-Estimation-Activity-Classification>